

RENANCE

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by Renance

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Paper 1 of 1 — EEC125 NOTE

Level ND1 • First Semester • Handout

ELECTRICAL
ENGINEERING
SCIENCE II (EEC 125)

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This lecture note is divided into three chapters. Chapter 1 introduces the concepts and laws associated with

magnetic fields and their application to magnetic circuits and materials. Chapter 2 concerns the principles

and laws governing electromagnetic induction and the concepts of self and mutual inductance. chapter 3

deals with the concepts, terms and definitions associated with alternating quantities. Although an A.C. can

have any waveshape, the most common waveform is a sinewave. For this reason, unless specified otherwise,

you may assume that sinusoidal waveforms are implied.

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CHAPTER 1

MAGNETIC FIELD AND CIRCUIT

On completion of this chapter you should be able to:

- i. Describe the forces of attraction and repulsion between magnetised bodies.
- ii. Understand the various magnetic properties and quantities, and use them to solve simple series and parallel magnetic circuit problems.
- iii. Appreciate the effect of magnetic hysteresis, and the properties of different types of magnetic material.

1.1 INTRODUCTION

A magnet is a material that has two poles. These are the north seeking pole and the south seeking pole (S-pole). A simple experiment with a pair of magnet reveals two important facts:

- i. Like magnetic poles repel one another
- ii. Unlike magnetic poles attract one another

The phenomena of magnetism and electromagnetism are dependent upon a certain property of the medium called its permeability.

Every medium is supposed to possess two permeabilities:

- (i) Absolute permeability (μ) and
- (ii) Relative permeability (μ_r).

For measuring relative permeability, vacuum or free space is chosen as the reference medium.

It is allotted an absolute permeability of $\mu_0 = 4\pi \times 10^{-7}$ henry/metre. Obviously, relative

permeability of vacuum with reference to itself is unity. Hence, for free space, absolute permeability $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ relative permeability $\mu_r = 1$. Now, take any medium other than vacuum. If its relative permeability, as compared to vacuum is μ_r , then its absolute permeability is $\mu = \mu_0 \mu_r \text{ H/m}$.

1.2 MAGNETIC FIELD

A permanent magnet is a piece of ferromagnetic material (such as iron, nickel or cobalt) which has properties of attracting other pieces of these materials. A permanent magnet will position itself in a north and south direction when freely suspended. The north-seeking end of the magnet is called the north pole, N, and the south-seeking end the south pole, S.

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The area around a magnet is called the magnetic field and it is in this area that the effects of the magnetic force produced by the magnet can be detected. A magnetic field cannot be seen, felt, smelt or heard and therefore is difficult to represent. Michael Faraday suggested that the magnetic field could be represented pictorially, by imagining the field to consist of lines of magnetic flux, which enables investigation of the distribution and density of the field to be carried out.

Figure 1

1.3 WEBER AND EWING'S MOLECULAR THEORY

The basic assumption of this theory is that molecules of all substances are inherently magnets in themselves, each having a N and S pole. In an unmagnetised state, it is supposed that these small molecular magnets lie in all sorts of haphazard manner forming more or less closed loops. According to the laws of attraction and repulsion, these closed magnetic circuits are satisfied internally, hence there is no resultant external magnetism exhibited by the iron bar. But when such an iron bar is placed in a magnetic field or under the influence of a magnetising force, then these molecular magnets start turning round their axes and orientate themselves more or less along straight lines parallel to the direction of the magnetising force. This linear arrangement of the molecular magnets results in N polarity at one end of the bar and S polarity at the other.

1.4 CURIE POINT

As a magnetic material is heated, its molecules vibrate more violently. As a consequence, individual molecular magnets get out of alignment as the temperature is increased, thereby reducing the magnetic strength of the magnetised substance.

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The temperature at which the vibrations of the molecular magnets become so random and out of alignment as to reduce the magnetic strength to zero is called Curie point. More accurately, it is that critical temperature above which ferromagnetic material becomes paramagnetic.

1.5 CLASSIFICATION OF MAGNETIC MATERIALS

Although all material have magnetic properties of some kind being either diamagnetic , paramagnetic or ferro-magnetic, the term magnetic material is customarily applied only to substances which exhibit ferro-magnetism.

Diamagnetism: is a weak form of magnetism which arises only when an external field is applied. It arises due to change in the orbital motion of electrons on application of a magnetic field. There are no magnetic dipoles in the absence of a magnetic field and when a magnetic field is applied the dipole moments are aligned opposite to field direction. The magnetic susceptibility is negative i.e. B in a diamagnetic material is less than that of vacuum. Examples are zinc, mercury, sulphur, copper, silver etc.

Paramagnetism: In a paramagnetic material the cancellation of magnetic moments between electron pairs is incomplete and hence magnetic moments exist without any external magnetic field. However, the magnetic moments are randomly aligned and hence no net magnetization without any external field. When a magnetic field is applied all the dipole moments are aligned in the direction of the field. The magnetic susceptibility is small but positive. i.e. B in a paramagnetic material is slightly greater than that of vacuum. Examples are tin, aluminium, magnesium, platinum, manganese etc.

Ferromagnetism: Certain materials possess permanent magnetic moments in the absence of an external magnetic field. This is known as ferromagnetism. Permanent magnetic moments in ferromagnetic materials arise due to uncanceled electron spins by virtue of their electron structure. The coupling interactions of electron spins of adjacent atoms cause alignment of moments with one another. Examples are iron, steel, nickel, cobalt and some of their alloys. Ferromagnetic materials are of two types: soft magnetic material and hard magnetic material

a. Soft magnetic materials do not retain their magnetism for any appreciable time after the magnetisation force has been removed. Soft magnets have a narrow hysteresis curve and high initial permeability and hence easy to magnetize and demagnetize. Soft magnets like Fe, Fe-Si are useful when rapid magnetization and demagnetization is required as in transformer cores.

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b. Hard magnetic materials retain their magnetism for a long time after the magnetisation force has been removed hence they are used for making permanent magnet. Hard magnets are used in all permanent magnets in applications such as power drills, motors, speakers.

1.6 LAW OF MAGNETIC FORCE

Coulomb was the first to determine experimentally the quantitative expression for the magnetic force between two isolated point poles. The law states that the force of attraction or repulsion between two magnetic poles is directly proportional to their pole strengths and inversely proportional to the square of the distance between them.

For example, if m_1 and m_2 represent the magnetic strength of the two poles, r the distance between.

$$\frac{1}{r^2}$$

$$\propto$$

$$\frac{1}{r^2}$$

$$\frac{1}{r^2}$$

$$\frac{1}{r^2}$$

$$\frac{1}{r^2}$$

In the S.I. system of units, the value of the constant k is $= \frac{1}{4\pi\mu_0\mu_r}$.

Where μ_0 = absolute permeability and μ_r = relative permeability

If, in the above equation, $m_1 = m_2 = m$ (say); $r = 1$ metre; and the medium is air

$$F = \frac{1}{4\pi\mu_0} N$$

Hence, a unit magnetic pole may be defined as that pole which when placed in vacuum at a distance of one metre from a similar and equal pole repels it with a force of $\frac{1}{4\pi\mu_0}$ newton.

1.7 MAGNETIC FIELD STRENGTH (H)

Magnetic field strength at any point within a magnetic field is numerically equally to the force experienced by a N-pole of one weber placed at that point. Hence, unit of H is N/Wb.

Magnetic field strength is also defined as the mmf per metre length of the magnetic circuit. The quantity symbol for magnetic field strength is H , the unit of measurement being ampere turn/metre.

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$$\frac{1}{r^2}$$

$$\frac{1}{r^2} = \text{ampere turn/metre}$$

$$\frac{1}{r^2}$$

Where l is the mean or average length of the magnetic circuit.

It would be helpful to remember that following terms are sometimes interchangeably used with field intensity: Magnetising force, strength of field, magnetic intensity and intensity of magnetic field.

1.8 MAGNETIC POTENTIAL

The magnetic potential at any point within a magnetic field is measured by the work done in shifting a N-pole of one weber from infinity to that point against the force of the magnetic field.

It is given by

It is a scalar quantity.

1.9 FLUX PER UNIT POLE

Magnetic flux is the amount of magnetic field (or the number of lines of force) produced by a magnetic source. A unit N-pole is supposed to radiate out a flux of one weber. Its symbol is Φ . Therefore, the flux coming out of a N-pole of m weber is given by

$$\Phi = m \frac{1}{r^2}$$

1.10 FLUX DENSITY (B)

It is given by the flux passing per unit area through a plane at right angles to the flux. It is usually designated by the capital letter B and is measured in weber/meter². It is a Vector Quantity. If Φ Wb is the total magnetic flux passing normally through an area of A m², then

$$B = \frac{\Phi}{A}$$

$$B = \frac{\Phi}{A} \text{ Wb/m}^2 \text{ (i)}$$

$$B = \mu H$$

1.11 ABSOLUTE PERMEABILITY (μ) AND RELATIVE PERMEABILITY (μ_r)

In Fig. 2 is shown a bar of a magnetic material, say, iron placed in a uniform field of strength H N/Wb. Suppose, a flux density of B Wb/m² is developed in the rod.

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Figure 2

Then, the absolute permeability of the material of the rod is defined as

$$\mu = B/H \text{ henry/metre or } B = \mu H = \mu_0 \mu_r H \text{ Wb/m}^2 \dots \dots \dots \text{ (i)}$$

When H is established in air (or vacuum), then corresponding flux density developed in air is

$$B_0 = \mu_0 H$$

Now, when iron rod is placed in the field, it gets magnetised by induction. If induced pole strength in the rod is m Wb, then a flux of m Wb emanates from its N-pole, re-enters its S-pole and continues from S to N-pole within the magnet. If A is the face or pole area of the magnetised iron bar, the induction flux density in the rod is

$$B_i = m/A \text{ Wb/m}^2$$

Hence, total flux density in the iron rod consists of two parts [Fig. 2 (b)].

(i) B_0 -flux density in air even when rod is not present

(ii) B_i -induction flux density in the rod

$$B = B_0 + B_i = \mu_0 H + m/A$$

Eq. (i) above may be written as $B = \mu_r \mu_0 H = \mu_r B_0$

$$\mu_r = \frac{B}{B_0}$$

$$\mu_r = \frac{B}{B_0}$$

$$\mu_r = \frac{B}{B_0}$$

Hence, relative permeability of a material is equal to the ratio of the flux density produced in that material to the flux density produced in vacuum by the same magnetising force.

1.12 INTENSITY OF MAGNETISATION (I)

It may be defined as the induced pole strength developed per unit area of the bar. Also, it is the magnetic moment developed per unit volume of the bar.

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Let m = pole strength induced in the bar in Wb

A = face or pole area of the bar in m²

Then $I = m/A \text{ Wb/m}^2$

Hence, it is seen that intensity of magnetisation of a substance may be defined as the flux density produced in it due to its own induced magnetism.

If l is the magnetic length of the bar, then the product $(m \times l)$ is known as its magnetic moment M .

$$M = m \times l$$

$$M = \frac{m}{A} \times l = \frac{M}{A} \times l$$

$$M = m \times l$$

1.13 SUSCEPTIBILITY (K)

Susceptibility is defined as the ratio of intensity of magnetisation I to the magnetising force H .

Therefore $K = I/H$ henry/metre.

1.14 Relation between B , H , I And K

It is obvious from the above discussion that flux density B in a material is given by

$$B = B_0 + \frac{m}{A} = B_0 + I \therefore B = \mu_0 H + I$$

Now absolute permeability is

$$\mu = \frac{B}{H} = \frac{B_0 + I}{H}$$

$$\mu = \frac{B_0}{H} + \frac{I}{H}$$

$$\mu = \mu_0 + \mu_r$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{\mu_0 + \mu_r \mu_0}{\mu_0} = 1 + \mu_r$$

Also

$$\mu = \mu_0 (1 + K)$$

$$\mu_r = 1 + K$$

$$\mu_0 (1 + K) = \mu_0 + \mu_r \mu_0 \therefore 1 + K = 1 + \mu_r$$

For ferro-magnetic and para-magnetic substances, K is positive and for diamagnetic substances, it is negative. For ferro-magnetic substance (like iron, nickel, cobalt and alloys like nickel-iron and cobalt-iron) μ_r is much greater than unity whereas for para-magnetic

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substances (like aluminium), μ_r is slightly greater than unity. For diamagnetic materials (bismuth) $\mu_r < 1$.

Example 1: The magnetic susceptibility of oxygen gas at 20°C is $167 \times 10^{-11} \text{ H/m}$. Calculate its absolute and relative permeability.

Solution

$$\mu = \mu_0 + \mu_r \mu_0$$

$$\mu = 4\pi \times 10^{-7} + 167 \times 10^{-11} = 1.258 \times 10^{-7} \text{ H/m}$$

$$\mu_r = \frac{\mu}{\mu_0} = \frac{1.258 \times 10^{-7}}{4\pi \times 10^{-7}}$$

$$\mu_r = 1$$

$$\mu_0 4\pi \times 10^{-7}$$

Example 2: The pole of a bar magnet measures 1.2cm x 2.1cm and the associated flux is 10mWb. Calculate the flux density.

Solution

$$\Phi = 10 \times 10^{-3}$$

$$B = \frac{\Phi}{A} = \frac{39.68 \text{ mWb}}{1.2 \times 10^{-2} \times 2.1 \times 10^{-2}}$$

$$B = 1.2 \times 10^{-2} \times 2.1 \times 10^{-2}$$

Example 3: Two magnetic north poles of strength 0.005 and 0.01 Wb are placed at a distance of 5cm in air. Determine the force between them with its nature.

Solution

$$\Phi_1 = 0.005$$

$$\Phi_2 = 0.01$$

$$r = 5 \text{ cm}$$

$$k = 10^7$$

$$F = \frac{k \Phi_1 \Phi_2}{r^2} = \frac{10^7 \times 0.005 \times 0.01}{(5 \times 10^{-2})^2}$$

$$F = \frac{4 \times 10^4 \times 4 \times 10^{-7} \times 1}{0.0025}$$

$$0.005 \times 0.01$$

$$F = 63325.74 \times 10^{-4} = 1266.51 \text{ N}$$

$$(5 \times 10^{-2})^2$$

1.15 MAGNETIC CIRCUIT

It may be defined as the route or path which is followed by magnetic flux. The law of magnetic circuit are quite similar to (but not the same as) those of the electric circuit. Consider a solenoid or a toroidal iron ring having a magnetic path of l metre, area of cross section A m² and a coil

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of N turns carrying I amperes wound anywhere on it as in Fig. 1 Field strength inside the solenoid is

Figure 3

$$H = \frac{NI}{l}$$

$$B = \mu_0 \mu_r H$$

$$\Phi = BA$$

$$\Phi_0 = \frac{NI^2 \mu_r \mu_0 A}{l}$$

$$\Phi = \frac{\Phi_0}{\mu_r \mu_0} = \frac{NI^2 \mu_r \mu_0 A}{l \mu_r \mu_0}$$

$$\Phi = \frac{NI^2 \mu_r \mu_0 A}{l \mu_r \mu_0}$$

$$\Phi_0 = \frac{NI^2 \mu_r \mu_0 A}{l}$$

$$\text{The total flux produce } \Phi = \frac{\Phi_0}{\mu_r \mu_0} = \frac{NI^2 \mu_r \mu_0 A}{l \mu_r \mu_0}$$

$$\Phi = \frac{NI^2 \mu_r \mu_0 A}{l \mu_r \mu_0}$$

$$\Phi = \frac{NI^2 \mu_r \mu_0 A}{l \mu_r \mu_0}$$

$$\Phi = \frac{NI}{\frac{l}{\mu_0 \mu_r}}$$

$$\frac{1}{\frac{l}{\mu_0 \mu_r}}$$

The numerator 'NI' which produces magnetization in the magnetic circuit is known as magnetomotive force (m.m.f.). Obviously, its unit is ampere-turn (AT). It is analogous to e.m.f. in an electric circuit.

The denominator $\frac{l}{\mu_0 \mu_r}$ is called the reluctance of the circuit and is analogous to resistance in electric circuits.

Therefore flux = m.m.f./reluctance or $\Phi = \frac{NI}{\frac{l}{\mu_0 \mu_r}}$

Sometimes, the above equation is called the "Ohm's Law of Magnetic Circuit" because it resembles a similar expression in electric circuits i.e.

current = e.m.f. / resistance or $I = V/R$

1.16 DEFINITIONS CONCERNING MAGNETIC CIRCUIT

1. Magnetomotive force (m.m.f.). It drives or tends to drive flux through a magnetic circuit and corresponds to electromotive force (e.m.f.) in an electric circuit. M.M.F. is

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equal to the work done in joules in carrying a unit magnetic pole once through the entire magnetic circuit. It is measured in ampere-turns.

$$F = \Phi S = NI$$

In fact, as p.d. between any two points is measured by the work done in carrying a unit charge from one point to another, similarly, m.m.f. between two points is measured by the work done in joules in carrying a unit magnetic pole from one point to another.

2. Ampere-turns (AT). It is the unit of magnetometre force (m.m.f.) and is given by the product of number of turns of a magnetic circuit and the current in amperes in those turns.

3. Reluctance. It is the name given to that property of a material which opposes the creation of magnetic flux in it. It, in fact, measures the opposition offered to the passage of magnetic flux through a material and is analogous to resistance in an electric circuit even in form. Its units is AT/Wb.

$$\frac{1}{\frac{l}{\mu_0 \mu_r}}$$

$$\frac{1}{\frac{l}{\mu_0 \mu_r}} =$$

$$\frac{1}{\frac{l}{\mu_0 \mu_r}}$$

In other words, the reluctance of a magnetic circuit is the number of amp-turns required per weber of magnetic flux in the circuit. Since $1 \text{ AT/Wb} = 1/\text{henry}$, the unit of reluctance is "reciprocal henry."

4. Permeance. It is reciprocal of reluctance and implies the ease or readiness with which magnetic flux is developed. It is analogous to conductance in electric circuits. It is measured in terms of Wb/AT or henry.

5. Reluctivity. It is specific reluctance and corresponds to resistivity which is 'specific

resistance’.

1.17 COMPARISON BETWEEN MAGNETIC AND ELECTRIC CIRCUITS

Similarities

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Differences

1. Strictly speaking, flux does not actually ‘flow’ in the sense in which an electric current flows.
2. If temperature is kept constant, then resistance of an electric circuit is constant and is independent of the current strength (or current density). On the other hand, the reluctance of a magnetic circuit does depend on flux (and hence flux density) established in it. It is so because μ (which equals the slope of B/H curve) is not constant even for a given material as it depends on the flux density B. Value of μ is large for low value of B and vice-versa. Hence, reluctance is small ($S = l/\mu A$) for small values of B and large for large values of B.
3. Flow of current in an electric circuit involves continuous expenditure of energy but in a magnetic circuit, energy is needed only creating the flux initially but not for maintaining it.

Example : A toroid of mean radius 40 mm, effective csa 3cm^2 , and relative permeability 150, is wound with a 900 turn coil that carries a current of 1.5 A. Calculate (a) the mmf, (b) the magnetic field strength and (c) the flux and flux density.

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Solution

$$\text{MMF} = NI = 900 \times 1.5 = 1350 \text{ AT}$$

$$H = \frac{\text{MMF}}{2\pi r}$$

$$H = \frac{1350}{2\pi \times 40 \times 10^{-3}} = 5371.48 \text{ AT/m}$$

$$\mu_r = 150$$

$$\mu = \mu_r \mu_0 = 150 \times 4\pi \times 10^{-7} \times 150 \times 5371.48 = 1.0125 \text{ T}$$

$$\Phi = BA = 1.0125 \times 3 \times 10^{-4} = 303.75 \mu\text{Wb}$$

Example: A steel toroid of the dimensions shown in Fig. example 5 is wound with a 500 turn coil of wire. What value of current needs to be passed through this coil in order to produce a flux of $250 \mu\text{Wb}$ in the toroid, if under these conditions the relative permeability of the toroid is 300. Ans: $I = 0.55 \text{ amp}$

Figure example 5

Example: A solenoid with a core of csa of 15 cm^2 and relative permeability 65, produces a flux of $200 \mu\text{Wb}$. If the core material is changed to one of relative permeability 800 what will be the new flux and flux density? Ans: $B = 1.641 \text{ T}$ and $\Phi = 2.462 \text{ mWb}$

Example : A mild steel ring having a csa of 5cm^2 and mean circumference of 40 cm has a

coil of 200 turns wound uniformly around it. Calculate (i) reluctance of the ring (ii) current required to produce a flux of $800 \mu\text{Wb}$ in the ring. Take relative permeability of mild steel as 380 and $\mu_0 = 4\pi \times 10^{-7}$ Ans: $S = 1.675 \times 10^6 \text{ AT/Wb}$ $I = 6.7 \text{ A}$

1.18 B-H CURVES (MAGNETIZATION CURVES)

For ferromagnetic materials, μ is not constant but varies with flux density and there is no easy way to

compute it. In reality, however, it isn't μ that you are interested in: What you really want to know is, given B, what is H, and vice versa. A set of curves, called B-H or magnetization curves, provides this information. (These curves are obtained experimentally and are available in handbooks. A separate curve is required for each material.) Figure shows typical curves for cast iron, cast steel, and sheet steel

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Figure 4

1.19 AMPERE'S CIRCUITAL LAW:

If we apply the analogy to Kirchhoff's voltage law (for magnetic circuits).

States that the algebraic sum of the rises and drops of the mmf around a closed loop of a magnetic circuit is equal to zero; that is, the sum of the rises in mmf equals the sum of the drops in mmf around a closed loop. Equation is referred to as Ampère's circuital law. When it is applied to magnetic circuits, sources of mmf are expressed by the equation. The summation is algebraic and terms are additive or subtractive, depending on the direction of flux and how the coils are wound.

Consider the magnetic circuit appearing in Figure constructed of three different ferromagnetic materials. Applying Ampère's circuital law, we have

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Which states that the applied mmf NI is equal to the sum of the drops HL around the loop. The path to use for the HL terms is the mean (average) path.

You now have two magnetic circuit models (Figure below). While the reluctance model (a) is not very useful for solving problems, it helps relate magnetic circuit problems to familiar electrical circuit concepts. The Ampere's law model, on the other hand, permits us to solve practical problems.

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1.20 COMPOSITE SERIES MAGNETIC CIRCUIT

Figure 5

In Fig. 5 is shown a composite series magnetic circuit consisting of three different magnetic materials of different permeability and lengths and one air gap ($\mu_r = 1$). Each path will

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have its own reluctance. The total reluctance is the sum of individual reluctances as they are joined in series.

□

□□□□□ □□□□□□□□□□ = Σ

□0 □□ □

□1 □2 □3 □□

= + + +

□0 □□ □1 □0 □□ □2 □0 □□ □3 □0 □□

□. □. □

□□□□ Φ =

□

□0 □□ □

How to Find Ampere-turns

□□□□□□_□□□□□ □□ = $\square \times \square$

Hence, following procedure should be adopted for calculating the total ampere turns of a composite magnetic path.

(i) Find H for each portion of the composite circuit. For air, $H = B/\mu_0$, otherwise $H = B/\mu_0\mu_r$.

(ii) Find ampere-turns for each path separately by using the relation $AT = H \times l$.

(iii) Add up these ampere-turns to get the total ampere-turns for the entire circuit.

Example: A series magnetic circuit comprises of three sections (i) length of 80 mm with cross-sectional area 60 mm², (ii) length of 70 mm with cross-sectional area 80 mm² and (iii) and airgap of length 0.5 mm with cross-sectional area of 60 mm². Sections (i) and (ii) are of a material having magnetic characteristics given by the following table.

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H (AT/m) 100 210 340 500 800 1500

B (Tesla) 0.2 0.4 0.6 0.8 1.0 1.2

Determine the current necessary in a coil of 4000 turns wound on section (ii) to produce a flux density of 0.8 Tesla in the air-gap. Neglect magnetic leakage.

Solution.

Section (i) It has the same cross-sectional area as the air-gap. Hence, it has the same flux density i.e. 0.8 Tesla as in the air-gap. The value of the magnetising force H corresponding to this flux density of 0.8T as read from the B-H plot is 500 AT/m.

m.m.f. reqd = $H \times l = 500 \times (80 \times 10^{-3}) = 40 \text{ AT}$

Section (ii) Since its cross-sectional area is different from that of the air-gap, its flux density would also be different even though, being a series circuit, its flux would be the same.

Air-gap flux = $B \times A = 0.8 \times (60 \times 10^{-6}) = 48 \times 10^{-6} \text{ Wb}$

Flux density in this section = $48 \times 10^{-6} / 80 \times 10^{-6} = 0.6 \text{ T}$

The corresponding value of the H from the given graph is 340 AT/m

m.m.f. reqd, for this section = $340 \times (70 \times 10^{-3}) = 23.8 \text{ AT}$.

Air-gap

$H = B/\mu_0 = 0.8/4\pi \times 10^{-7} = 0.6366 \times 10^6 \text{ AT/m}$

$\therefore \text{m.m.f. reqd.} = 0.6366 \times 10^6 \times (0.5 \times 10^{-3}) = 313.31 \text{ AT}$

Total m.m.f. reqd. = $40 + 23.8 + 313.31 = 377.11$

$NI = 377.11$ or $I = 377.11/4000 = 0.094 \text{ A}$

1.21 PARALLEL MAGNETIC CIRCUITS

Figure 6

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Fig. 6 (a) shown a parallel magnetic circuit consisting of two parallel magnetic paths ACB and ADB acted upon by the same m.m.f. Each magnetic path has an average length of $2(l_1 + l_2)$. The flux produced by the coil wound on the central core is divided equally at point A between the two outer parallel paths. The reluctance offered by the two parallel paths is = half the reluctance of each path.

Fig. 6 (b) shows the equivalent electrical circuit where resistance offered to the voltage source is $= R//R = R/2$

1.22 SERIES-PARALLEL MAGNETIC CIRCUITS

Figure 7

Such a circuit is shown in Fig. 7 (a). It shows two parallel magnetic circuits ACB and ACD connected across the common magnetic path AB which contains an air-gap of length l_g . As usual, the flux Φ in the common core is divided equally at point A between the two parallel paths which have equal reluctance. The reluctance of the path AB consists of

- (i) air gap reluctance and
- (ii) the reluctance of the central core which comparatively negligible.

Hence, the reluctance of the central core AB equals only the air-gap reluctance across which are connected two equal parallel reluctances. Hence, the m.m.f. required for this circuit would be the sum of

- (i) that required for the air-gap and
- (ii) that required for either of two paths (not both).

The equivalent electrical circuit is shown in Fig. 7 (b) where the total resistance offered to the voltage source is $= R_1 + R//R = R_1 + R/2$.

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Example: A magnetic circuit made of mild steel is arranged as shown in Fig. 8 The central

limb is wound with 500 turns and has a cross-sectional area of 800 mm². Each of the outer limbs has a cross-sectional area of 500 mm². The air-gap has a length of 1 mm. Calculate the current required to set up a flux of 1.3 mWb in the central limb assuming no magnetic leakage and fringing. Mild steel required 3800 AT/m to produce flux density of 1.625 T and 850 AT/m to produce flux density of 1.3 T.

Figure 8

Solution.

Flux in the central limb is = 1.3 mWb = 1.3×10^{-3} Wb

Cross section A = 800 mm² = 800×10^{-6} m²

$\therefore B = \Phi/A = 1.3 \times 10^{-3} / 800 \times 10^{-6} = 1.625$ T

Corresponding value of H for this flux density is given as 3800 AT/m.

Since the length of the central limb is 120 mm.

m.m.f. required is AT = $H \times l = 3800 \times (120 \times 10^{-3}) = 456$ AT/m.

Air-gap

Flux density in the air-gap is the same as that in the central limb.

$H = B/\mu_0 = 1.625/4\pi \times 10^{-7} = 0.1293 \times 10^7$ AT/m

Length of the air-gap = 1 mm = 10^{-3} m

m.m.f. reqd. for the air-gap = $H \times l = 0.1293 \times 10^7 \times 10^{-3} = 1293$ AT.

The flux of the central limb divides equally at point A in figure along the two parallel path ABCD and AFED. We may consider either path, say ABCD and calculate the m.m.f. required for it. The same m.m.f. will also send the flux through the other parallel path AFED.

Flux through ABCD = $1.3 \times 10^{-3}/2 = 0.65 \times 10^{-3}$ Wb

Flux density B = $0.65 \times 10^{-3} / 500 \times 10^{-6} = 1.3$ T

The corresponding value of H for this value of B is given at 850 AT/m.

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$\therefore$ m.m.f. reqd. for path ABCD = $H \times l = 850 \times (300 \times 10^{-3}) = 255$ AT

As said above, this, m.m.f. will also send the flux in the parallel path AFED.

Total m.m.f. reqd. = $456 + 1293 + 255 = 2004$ AT

Since the number of turns is 500,

$I = 2004/500 = 4$ A.

Example: A cast steel D.C. electromagnet shown in Fig. 9 has a coil of 1000 turns on its central limb. Determine the current that the coil should carry to produce a flux of 2.5 mWb in the air-gap. Neglect leakage. Dimensions are given in cm. The magnetisation curve for cast steel is as under:

Flux density (Wb/m²): 0.2 0.5 0.7 1.0 1.2

Amp-turns/metre: 300 540 650 900 1150

Figure 9

Solution.

Two points should be noted (i) there are two (equal) parallel paths ACDE and AGE across the

central path AE. (ii) flux density in either parallel path is half of that in the central path because flux divides into two equal parts at point A. Total m.m.f. required for the whole electromagnet is equal to the sum of the following three m.m.fs.

i. that required for path EF

ii. that required for air-gap

iii. that required for either of the two parallel paths; say, path ACDE

Flux density in the central limb and air gap is

$$= 2.5 \times 10^{-3} / (5 \times 5) \times 10^{-4} = 1 \text{ Wb/m}^2$$

Corresponding value of H as found from the given data is 900 AT/m.

$$\therefore \text{AT for central limb} = 900 \times 0.3 = 270$$

$$H \text{ in air-gap} = B/\mu_0 = 1/4\pi \times 10^{-7} = 79.56 \times 10^4 \text{ AT/m}$$

$$\text{AT required} = 79.56 \times 10^4 \times 10^{-3} = 795.6$$

Flux density in path ACDE is 0.5 Wb/m² for which corresponding value of H is 540 AT/m.

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$$\therefore \text{AT required for path ACDE} = 540 \times 0.6 = 324$$

$$\text{Total AT required} = 270 + 795.6 + 324 = 1390$$

$$\text{Current required} = 1390/1000 = 1.39 \text{ A}$$

1.23 MAGNETIC HYSTERESIS

Hysteresis comes from a Greek word meaning 'to lag behind'. It is found that when the magnetic field strength in a magnetic material is varied, the resulting flux density displays a lagging effect.

Consider such a specimen of magnetic material that initially is completely unmagnetised. If no current flows through the magnetising coil then both H and B will initially be zero. The value of H is now increased by increasing the coil current in discrete steps. The corresponding flux density is then noted at each step. If these values are plotted on a graph until magnetic saturation is achieved, the dotted curve (the initial magnetisation curve) shown in Fig. 10 results.

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Figure 10

Let the current now be reduced (in steps) to zero, and the corresponding values for B again noted and plotted. This would result in the section of graph from A to C. This shows that when the current is zero once more (so $H = 0$), the flux density has not reduced to zero. The flux density remaining is called the remanent flux density (OC). This property of a magnetic material, to retain some flux after the magnetising current is removed, is known as the remanence or retentivity of the material.

Let the current now be reversed, and increased in the opposite direction. This will have the effect of opposing the residual flux. Hence, the latter will be reduced until at some value of -H it reaches zero (point D on the graph). The amount of reverse magnetic field strength required to reduce the residual flux to zero is known as the coercive force. This property of a material

is called its coercivity.

If we now continue to increase the current in this reverse direction, the material will once more reach saturation (at point E). In this case it will be of the opposite polarity to that achieved at point A on the graph.

Once again, the current may be reduced to zero, reversed, and then increased in the original direction. This will take the graph from point E back to A, passing through points F and G on the way.

Note that residual flux density shown as OC has the same value, but opposite polarity, to that shown as OF. Similarly, coercive force OD = OG.

In taking the specimen through the loop ACDEFGA we have taken it through one complete magnetisation cycle. The loop is referred to as the hysteresis loop. The degree to which a

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material is magnetised depends upon the extent to which the 'molecular magnets' have been aligned.

Thus, in taking the specimen through a magnetisation cycle, energy must be expended. This energy is proportional to the area enclosed by the loop, and the rate (frequency) at which the cycle is repeated.

Magnetic materials may be subdivided into what are known as 'hard' and 'soft' magnetic materials. A hard-magnetic material is one which possesses a large remanance and coercivity. It is therefore one which retains most of its magnetism, when the magnetising current is removed. It is also difficult to demagnetise. These are the materials used to form permanent magnets, and they will have a very 'fat' loop as illustrated in Fig. 11(a). A soft magnetic material, such as soft iron and mild steel, retains very little of the induced magnetism. It will therefore have a relatively 'thin' hysteresis loop, as shown in Fig. 11 (b). The soft magnetic materials are the ones used most often for engineering applications. Examples are the magnetic circuits for rotating electric machines (motors and generators), relays, and the cores for inductors and transformers.

Figure 11

When a magnetic circuit is subjected to continuous cycling through the loop a considerable amount of energy is dissipated. This energy appears as heat in the material. Since this is normally an undesirable effect, the energy thus dissipated is called the hysteresis loss. Thus, the thinner the loop, the less wasted energy. This is why 'soft' magnetic materials are used for the applications listed above.

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CHAPTER 2

ELECTROMAGNETISM

This chapter concerns the principles and laws governing electromagnetic induction and the concepts of self and mutual inductance.

On completion of this chapter you should be able to use these principles to:

- i. Understand the basic operating principles of motors and generators.
- ii. Carry out simple calculations involving the generation of voltage, and the production of force and torque.
- iii. Determine the value of inductors, and apply the concepts of self and mutual inductance to the operating principles of transformers.

2.1 INTRODUCTION

It is well known that whenever an electric current flows through a conductor, a magnetic field is immediately brought into existence in the space surrounding the conductor. It can be said that when electrons are in motion, they produce a magnetic field. The converse of this is also true i.e. when a magnetic field embracing a conductor moves relative to the conductor, it produces a flow of electrons in the conductor. This phenomenon whereby an e.m.f. and hence current (i.e. flow of electrons) is induced in any conductor which is cut across or is cut by a magnetic flux is known as electromagnetic induction.

Most applications of magnetism involve magnetic effects due to electric currents. The current, I , creates a magnetic field that is concentric about the conductor, uniform along its length, and whose strength is directly proportional to I .

2.2 FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

It is mainly due to the pioneering work of Michael Faraday, in the nineteenth century, that the modern technological world exists as we know it. Without the development of the generation of electrical power, such advances would have been impossible. Thus, although the concepts involved with electromagnetic induction are very simple, they have far-reaching influence. Faraday's law is best considered in two interrelated parts:

- 1 The value of emf induced in a circuit or conductor is directly proportional to the rate of change of magnetic flux linking with it.
- 2 The polarity of such an emf, induced by an increasing flux, is opposite to that induced by a decreasing flux.

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Figure 1

$$- \frac{d\Phi}{dt}$$

$$e = - \frac{d\Phi}{dt} \quad \text{--- (1)}$$

Φ

Equation 1 forms the basis for the definition of the unit of magnetic flux, the weber, thus:

The weber is that magnetic flux which, linking a circuit of one turn, induces in it an emf of one volt when the flux is reduced to zero at a uniform rate in one second. In other words, 1 volt = 1 weber/second or 1 weber = 1 volt second.

2.3 Lenz's Law

This law states that the polarity of an induced emf is always such that it opposes the change which produced it. This is similar to the statement in mechanics that for every force there is an

opposite reaction.

2.4 Fleming's Righthand Rule

This is a convenient means of determining the polarity of an induced emf in a conductor. Also, provided that the conductor forms part of a complete circuit, it will indicate the direction of the resulting current flow.

The first finger, the second finger and the thumb of the righthand are held out mutually at right angles to each other (like the three edges of a cube as shown in Fig 2a). The First finger indicates the direction of the Flux, the thuMb the direction of Motion of the conductor relative to the flux, and the sECond finger indicates the polarity of the induced Emf, and direction of Current flow. This process is illustrated in Fig. 2b, which shows the cross-section of a conductor being moved vertically upwards at a constant velocity through the magnetic field.

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(a) (b)

Figure 2

Example 1: The flux linking a 100 turn coil changes from 5mWb to 10mWb in a time of 5ms. Calculate the average emf induced in the coil.

Example 2: A 250 turn coil is linked by a magnetic flux that varies as follows: an increase from zero to 20mWb in a time of 0.05s; constant at this value for 0.02s; followed by a decrease to 4mWb in a time of 0.01s. Assuming that these changes are uniform, draw a sketch graph (i.e. not to an accurate scale) of the variation of the flux and the corresponding emf induced in the coil, showing all principal values.

2.5 EMF INDUCED IN A SINGLE STRAIGHT CONDUCTOR

Consider a conductor moving at a constant velocity v metre per second at right angles to a magnetic field having the dimensions shown in Fig. 3. The direction of the induced emf may be obtained using Fleming's righthand rule, and is shown in the diagram. Equation 1 is applicable, and in this case, the value for N is 1.

Figure 3

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Thus

$d\Phi$

$\square =$

$\square \square$

But flux is constant

Φ

$\square =$

$\square$

$$\Phi = BA$$

BA

$$B = \frac{\Phi}{A}$$

A

The cross sectional area of the field $A = l \times b$

Bld

$$B = \frac{\Phi}{l \times b}$$

l

b

$$B = \frac{\Phi}{l \times b} \quad \text{---(2)}$$

l

The above equation is only true for the case when the conductor is moving at right angles to the magnetic field. If the conductor moves through the field at some angle less than 90° , then the 'cutting' action between the conductor and the flux is reduced. This results in a consequent reduction in the induced emf. Thus, if the conductor is moved horizontally through the field, the 'cutting' action is zero, and so no emf is induced. To be more precise, we can say that only the component of the velocity at 90° to the flux is responsible for the induced emf. In general therefore, the induced emf is given by

$$e = B l v \sin \theta$$

where $v \sin \theta$ is the component of velocity at 90° to the field, as illustrated in Fig. 4 .

Figure 4

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Example 3: A conductor is moved at a velocity of 5m/s at an angle of 60° to a uniform magnetic field of 1.6mWb. The field is produced by a pair of pole pieces, the faces of which measure 10cm by 4cm. If the conductor length is parallel to the longer side of the field, calculate the emf induced.

This section, covering the induction or generation of an emf in a conductor moving through a magnetic field, forms the basis of the generator principle. However, most electrical generators are rotating machines, and we have so far considered only linear motion of the conductor.

Consider the conductor now formed into the shape of a rectangular loop, mounted on to an axle. This arrangement is then rotated between the poles of a permanent magnet. We now have the basis of a simple generator as illustrated in Fig. 5

Figure 5

The two sides of the loop that are parallel to the pole faces will each have an effective length l metre. At any instant of time, these sides are passing through the field in opposite directions. Applying the right hand rule at the instant shown in Fig. 5, the directions of the induced emfs will be as marked, i.e. of opposite polarities. However, if we trace the path around the loop, it will be seen that both emfs are causing current to flow in the same direction around the loop.

This is equivalent to two cells connected in series as shown in Fig. 6

The situation shown in Fig. 5 applies only to one instant in one revolution of the loop (it is equivalent to a 'snapshot' at that instant).

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Figure 6

If we were to plot a graph of the total emf generated in the loop, for one complete revolution, it would be found to be one cycle of a sinewave, i.e. an alternating voltage.

2.6 FORCE ON A CURRENT-CARRYING CONDUCTOR

Figure 7(a) shows the field patterns produced by two pole pieces, and the current flowing through the conductor. Since the lines of flux obey the rule that they will not intersect, then the flux pattern from the poles will be distorted as illustrated in Fig. 7(b). Also, since the lines of flux tend to act as if elastic, then they will try to straighten themselves. This results in a force being exerted on the conductor, in the direction shown.

Figure 7

The direction of this force may be more simply obtained by applying Fleming's left hand rule. This rule is similar to the right hand rule. The major difference is of course that the fingers and thumb of the left hand are now used. In this case, the First finger indicates the direction of the main Flux (from the poles). The seCond finger indicates the direction of Current flow. The thuMb shows the direction of the resulting force and hence consequent Motion. This is shown in Fig. 8

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Figure 8

We can derive an expression for the magnetic force on a current by taking a sum of the magnetic forces on individual charges.

The force on an individual charge moving at the drift velocity V_d is given by

$$F = qvB \sin \theta$$

$$F = NqvB \sin \theta \text{ where } N \text{ is number of charge carriers in the section of the wire of length } l$$

$$F = nAqvB \sin \theta \text{ where } n = \text{number of charge carriers per unit volume}$$

$$F = (nAqV_d)lB \sin \theta$$

$$ILB \sin \theta = F$$

$$F = ILB \sin \theta \text{ --- (3)}$$

Any conductor extending beyond the main field does not contribute to the force exerted.

Example 4: A pair of pole pieces 5cm by 3cm produce a flux of 2.5mWb. A conductor is placed in this field with its length parallel to the longer dimension of the poles. When a current is passed through the conductor, a force of 1.25N is exerted on it. Determine the value of the current.

If the conductor was placed at 45° to the field, what then would be the force exerted?

2.6 FORCE BETWEEN PARALLEL CONDUCTORS

When two parallel conductors are both carrying current their magnetic fields will interact to produce a force of attraction or repulsion between them. This is illustrated in Fig. 9

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Figure 9

In order to determine the value of such a force, consider first a single conductor carrying a current of I ampere. The magnetic field produced at some distance d from its centre is shown in Fig. 10.

Figure 10

□□

In general, $B = \frac{\mu NI}{2\pi d}$

But in this case, $N = 1$ and $I = 2\pi d$, so

□□

□=

2□□

Now, flux density $B = 0.00125$ tesla, and as the field exists in air, then $\mu_r = 1$. Thus, the flux density at distance d from the centre is given by

□□ □

□=

2□□

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Consider now two conductors Y and Z carrying currents I_1 and I_2 respectively, at a distance of d metres between their centres as in Fig 11

Figure 11

we can say that the flux density acting on Z due to current I_1 flowing in Y is:

□□ □1

□=

2□□

And the force exerted on Z = $B_1 I_2 l$ newton, or $B_1 I_2$ newton per meter length of Z

Hence force/meter length acting on Z

□□ □1 □2

2□□

$4 \times 10^{-7} \times 1 \times 2$

2□□

$$2 \times 10^{-7} \text{ N/A}^2$$

□

Now, the current I_2 flowing in Z also produces a magnetic field which will exert a force on Y. Using the same reasoning as above, it can be shown that:

force/metre length acting on Y

$$2 \times 10^{-7} \text{ N/A}^2$$

□

If $I_1 = I_2 = 1\text{A}$ and $d = 1\text{m}$

Force exerted on each conductor $= 2 \times 10^{-7} \text{ N}$

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This value of force forms the basis for the definition of the ampere, namely: that current, which when maintained in each of two infinitely long parallel conductors situated in vacuum, and separated one metre between centres, produces a force of $2 \times 10^{-7} \text{ N}$ per metre length on each conductor.

Example 5: Two long parallel conductors are spaced 35mm between centres. Calculate the force exerted between them when the currents carried are 50 A and 40 A respectively.

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2.7 SELF AND MUTUAL INDUCTANCE

The effects of self and mutual inductance can be demonstrated by another simple experiment. Consider two coils, as shown in Figure 1. Coil 1 can be connected to a battery via a switch. Coil 2 is placed close to coil 1, but is not electrically connected to it. Coil 2 has a galvo connected to its terminals.

Figure 1

When the switch is closed, the current in coil 1 will rapidly increase from zero to some steady value. Hence, the flux produced by coil 1 will also increase from zero to a steady value. This changing flux links with the turns of coil 2, and therefore induces an emf into it. This will be indicated by a momentary deflection of the galvo pointer.

Similarly, when the switch is subsequently opened, the flux produced by coil 1 will collapse to zero. The galvo will again indicate that a momentary emf is induced in coil 2, but of the opposite polarity to the first case. Thus, an emf has been induced into coil 2, by a changing current (and flux) in coil 1. This is known as a mutually induced emf.

If the changing flux can link with coil 2, then it must also link with the turns of coil 1. Thus, there must also be a momentary emf induced in this coil. This is known as a self-induced emf. Any induced emf obeys Lenz's law. This self-induced emf must therefore be of the opposite polarity to the battery emf. For this reason, it is also referred to as a back emf. Unfortunately, it is extremely difficult to demonstrate the existence of this back emf. If a voltmeter was

connected across coil 1, it would merely indicate the terminal voltage of the battery.

2.8 SELF-INDUCTANCE

Self-inductance is that property of a circuit or component which causes a self-induced emf to be produced, when the current through it changes. The unit of self-inductance is the henry, which is defined as follows:

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A circuit has a self-inductance of one henry (1 H) if an emf of one volt is induced in it, when the circuit current changes at the rate of one ampere per second (1 A/s).

The quantity symbol for self-inductance is L. From the above definition, we can state the following equation

$$-e = L \frac{di}{dt}$$

$$e = L \frac{di}{dt}$$

$$\frac{e}{L} = \frac{di}{dt}$$

$$\int \frac{e}{L} dt = \int di$$

Example: A coil has a self-inductance of 0.25 H. Calculate the value of emf induced, if the current through it changes from 100 mA to 350 mA, in a time of 25ms.

Example Calculate the inductance of a circuit in which an emf of 30 V is induced, when the circuit current changes at the rate of 200 A/s.

2.9 SELF-INDUCTANCE AND FLUX LINKAGES

Consider a coil of N turns, carrying a current of I amp. Let us assume that this current produces a flux of ϕ weber. If the current now changes at a uniform rate of di/dt ampere per second, it will cause a corresponding change of flux of $d\phi/dt$ weber per second. Let us also assume that the coil has a self-inductance of L henry. The self-induced emf may be determined from

$$e = -L \frac{di}{dt}$$

$$e = -L \frac{di}{dt}$$

$$\frac{e}{L} = -\frac{di}{dt}$$

However, the induced emf is basically due to the rate of change of flux linkages

$$e = -N \frac{d\phi}{dt}$$

$$e = -N \frac{d\phi}{dt}$$

$$\frac{e}{N} = -\frac{d\phi}{dt}$$

Since both equations represent the same induced emf, then

$$L \frac{di}{dt} = N \frac{d\phi}{dt}$$

$$L = N \frac{d\phi}{di}$$

$$L = N \frac{d\phi}{di}$$

$$L = N \frac{d\phi}{di}$$

$$L = N \frac{d\phi}{di}$$

$$L = N \frac{d\phi}{di}$$

A coil which is designed to have a specific value of self-inductance is known as an inductor

If the current I through an N turn coil produces a flux of ϕ weber, then its self-inductance is given by the equation

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$\square \Phi$

$$\square = \square\square\square\square\square$$

1

Example: A coil of 150 turns carries a current of 10 A. This current produces a magnetic flux of 0.01 Wb. Calculate (a) the inductance of the coil, and (b) the emf induced when the current is uniformly reversed in a time of 0.1 s.

Example: A current of 8 A, when flowing through a 3000 turn coil, produces a flux of 4 mWb. If the current is reduced to 2 A in a time of 100 ms, calculate the emf thus induced in the coil. Assume that the flux is directly proportional to the current.

2.10 FACTORS AFFECTING INDUCTANCE

Consider a coil of N turns wound on to a non-magnetic core, of uniform cross sectional area A metre² and mean length L metre. The coil carries a current of I amp, which produces a flux of ϕ weber. We know that the inductance will be

$\square \Phi$

$$\phi = h\alpha\alpha\alpha\alpha, \text{ But } \phi = BA$$

1

□BA

Therefore $\square =$

1

□ I HI

$$\square\square\square\square\square = , \square\square\square =$$

11

□ 2 BA

$\square =$

11

1

$$h_{\text{total}} = 0$$

1

0 0 2 A

$\square =$

1

1

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} \begin{array}{|c|} \hline \square \\ \hline \end{array} \begin{array}{|c|} \hline \square \\ \hline \end{array} \begin{array}{|c|} \hline \square \\ \hline \end{array} =$$

0

$$\Phi_2$$

$$\Phi = \frac{NI}{\frac{l}{\mu_r \mu_0}}$$

$$\Phi$$

Example: A 600 turn coil is wound on to a non-magnetic core of effective length 45 mm and csa 4 cm². (a) Calculate the inductance (b) The number of turns is increased to 900. Calculate

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the inductance value now produced. (c) The core of the 900 turn coil is now replaced by an iron core having a relative permeability of 75, and of the same dimensions as the original.

Calculate the inductance in this case

2.11 MUTUAL INDUCTANCE

When a changing current in one circuit induces an emf in another separate circuit, then the two circuits are said to possess mutual inductance. The unit of mutual inductance is the henry, and is defined as follows. Two circuits have a mutual inductance of one henry, if the emf induced in one circuit is one volt, when the current in the other is changing at the rate of one ampere per second. The quantity symbol for mutual inductance is M, and expressing the above definition as an equation we have

$$\text{Induced emf in coil 2} = -M \frac{dI_1}{dt}$$

$$E_2 =$$

$$\text{Induced emf in coil 1} = -M \frac{dI_2}{dt}$$

$$-E_1 =$$

$$E_1 =$$

$$M \frac{dI_2}{dt}$$

$$E_1 =$$

$$-M \frac{dI_2}{dt}$$

$$E_2 = M \frac{dI_1}{dt}$$

$$E_1 =$$

This emf may also be expressed in terms of the flux linking coil 2. If all of the flux from coil 1 links with coil 2, then we have what is called 100% flux linkage. In practice, it is more usual for only a proportion of the flux from coil 1 to link with coil 2. Thus the flux linkage is usually less than 100%. This is indicated by a factor, known as the coupling factor, k. 100% coupling is indicated by $k = 1$. If there is no flux linkage with coil 2, then k will have a value of zero. So if zero emf is induced in coil 2, the mutual inductance will also be zero. Thus, the possible values for the coupling factor k, lie between zero and 1. Expressed mathematically, this is written as

$$0 \leq k \leq 1$$

Consider two coils possessing mutual inductance, and with a coupling factor less than 1. Let a change of current dI_1/dt amp/s in coil 1 produce a change of flux $d\phi_1/dt$ weber/s. The proportion of this flux change linking coil 2 will be $d\phi_2/dt$ weber/s. If the number of turns on

coil 2 is N_2 , then

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$$-N_2 \frac{d\Phi_2}{dt}$$

$$N_2 = \frac{\Phi_2}{\Phi_1}$$

$$\Phi_1$$

$$\frac{N_2}{N_1} \Phi_2 = \Phi_1$$

$$\Phi_1 = \frac{N_2}{N_1} \Phi_2$$

$$N_2 \Phi_2$$

$$\Phi = \frac{N_1 \Phi_1}{N_2}$$

$$\Phi_1$$

As with self-inductance for a single coil, mutual inductance is a property of a pair of coils. They therefore retain this property, regardless of whether or not an emf is induced. Hence, equation may be modified to

$$N_2 \Phi_2$$

$$\Phi =$$

$$\Phi_1$$

Example: Two coils, A and B, have 2000 turns and 1500 turns respectively. A current of 0.5 A, flowing in A, produces a flux of 60Wb. The flux linking with B is 83% of this value. Determine (a) the self-inductance of coil A, and (b) the mutual inductance of the two coils.

2.12 RELATIONSHIP BETWEEN SELF- AND MUTUAL-INDUCTANCE

Consider two coils of N_1 and N_2 turns respectively, wound on to a common non-magnetic core. If the reluctance of the core is S ampere turns/weber, and the coupling coefficient is unity, then

$$N_1^2 \Phi_1 = N_2^2 \Phi_2$$

$$N_1 = \frac{N_2}{\Phi_1} \Phi_2 =$$

$$\Phi_1$$

Therefore

$$N_1^2 \Phi_1 = N_2^2 \Phi_2$$

$$N_1 \Phi_2 = \frac{N_2}{N_1} \Phi_1 \quad (1)$$

$$\Phi_2$$

$$N_2 \Phi_1 = N_1 \Phi_2$$

$$\Phi = \frac{N_1 \Phi_1}{N_2}, \text{ then } \frac{N_1 \Phi_1}{N_2} = \frac{N_2 \Phi_2}{N_1}$$

$$N_1 \Phi_1 = N_2 \Phi_2$$

$$N_1 \Phi_2 = N_2 \Phi_1$$

$$\Phi =$$

$$N_1 \Phi_1 = N_2 \Phi_2$$

$$N_1 \Phi_1 = N_2 \Phi_2$$

$$N_1 \Phi_1 = N_2 \Phi_2$$

$$\Phi_1 \Phi_2$$

$$\Phi =$$

$$\Phi$$

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$$2$$

$$\Phi_{12} \Phi_{22}$$

$$\Phi = \dots \dots \dots (2)$$

$$\Phi^2$$

Comparing equation 1 and 2

$$\Phi^2 = \Phi_1 \Phi_2$$

Therefore

$$\Phi = \sqrt{\Phi_1 \Phi_2}$$

The above equation is correct only provided that there is 100% coupling between the coils; i.e. $k = 1$. If k is less than 1, then the general form of the equation, shown below

$$\Phi = k\sqrt{\Phi_1 \Phi_2}$$

Example: A 400 turn coil is wound onto a cast steel toroid having an effective length of 25 cm and csa 4.5 cm². If the steel has a relative permeability of 180 under the operating conditions, calculate the self-inductance of the coil.

2.13 INDUCTANCES IN SERIES

Let the two coils be so joined in series that their fluxes (or m.m.fs) are additive i.e., in the same direction.

Figure 2

Let M = coefficient of mutual inductance

L_1 = coefficient of self-inductance of 1st coil

L_2 = coefficient of self-inductance of 2nd coil.

$$\Phi \Phi$$

Then, self induced e.m.f. in A is $\mathcal{E}_1 = - \frac{d\Phi_1}{dt}$

$$\Phi \Phi$$

Mutually-induced e.m.f. in A due to change of current in B is $\mathcal{E}'_1 = - \frac{d\Phi_{12}}{dt}$

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$$\Phi \Phi$$

Self-induced e.m.f. in B is $\mathcal{E}_2 = - \frac{d\Phi_2}{dt}$

$$\Phi \Phi$$

Mutually-induced e.m.f. in B due to change of current in A is $\mathcal{E}'_2 = - \frac{d\Phi_{21}}{dt}$

$$\Phi \Phi$$

Total induced e.m.f. in the combination $= - \frac{d}{dt} (\Phi_1 + \Phi_2 + 2\Phi_{12})$

If L is the equivalent inductance then total induced e.m.f. in that single coil would have been

$$\mathcal{E}$$

$$= -\frac{d\Phi}{dt}$$

$$\Phi = \Phi_1 + \Phi_2 + 2\Phi_m$$

When the coils are so joined that their fluxes are in opposite directions

Figure 3

$$\Phi_1 = -\Phi_2, \mathcal{E}_1' = \mathcal{E}_2, \Phi_2 = -\Phi_1, \mathcal{E}_2' = \mathcal{E}_1$$

$$\Phi = \Phi_1 + \Phi_2 - 2\Phi_m$$

$$\mathcal{E}$$

$$\text{Total induced e.m.f.} = -\frac{d\Phi}{dt} (\Phi_1 + \Phi_2 - 2\Phi_m)$$

$$\Phi = \Phi_1 + \Phi_2 - 2\Phi_m$$

Example: Two coils with a coefficient of coupling of 0.5 between them, are connected in series so as to magnetise (a) in the same direction (b) in the opposite direction. The corresponding values of total inductances are for (a) 1.9 H and for (b) 0.7 H. Find the self-inductances of the two coils and the mutual inductance between them.

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2.14 INDUCTANCE IN PARALLEL

Figure 4

In Fig. 4, two inductances of values L_1 and L_2 henry are connected in parallel. Let the coefficient of mutual inductance between the two be M . Let i be the main supply current and i_1 and i_2 be the branch currents Obviously, $i = i_1 + i_2$

$$\mathcal{E}_1 = \mathcal{E}_2$$

$$= +$$

$$\mathcal{E}_1 = \mathcal{E}_2$$

In each coil, both self and mutually induced e.m.fs. are produced. Since the coils are in parallel, these e.m.fs. are equal. For a case when self-induced e.m.f., we get

$$\mathcal{E}_1 = \mathcal{E}_2 = \mathcal{E}_1 + \mathcal{E}_2$$

$$\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2 = \mathcal{E}_1 + \mathcal{E}_2$$

$$\mathcal{E}_1 = \mathcal{E}_2$$

Where L is the equivalent inductance of the two inductor

Hence

$$\mathcal{E}_1 = \mathcal{E}_2 = \mathcal{E}_1 + \mathcal{E}_2$$

$$\mathcal{E}_1 + \mathcal{E}_2 = \mathcal{E}_2 + \mathcal{E}_1$$

$$\mathcal{E}_1 = \mathcal{E}_2$$

$$\mathcal{E}_1 = \mathcal{E}_2$$

$$(\mathcal{E}_1 - \mathcal{E}_2) = (\mathcal{E}_2 - \mathcal{E}_1)$$

$$\mathcal{E}_1 = \mathcal{E}_2$$

$$L_1 (L_2 - M) L_2$$

$$L_1 (L_1 - M) L_1$$

Therefore

$$L_1 (L_2 - M) L_2 L_2$$

$$= +$$

$$L_1 (L_1 - M) L_1 L_1$$

$$L_1 (L_2 - M) L_2$$

$$= (L_1 + L_2) - 2M - (1)$$

$$L_1 (L_1 - M) L_1$$

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Also

$$L_1 L_1 L_2$$

$$L = L_1 + L_2$$

$$L_1 L_1 L_1$$

$$L_1 L_1 L_2$$

$$= (L_1 + L_2)$$

$$L_1 L_1 L_1$$

$$L_1 L_1 (L_2 - M) L_2 L_2$$

$$= (L_1 + L_2)$$

$$L_1 L_1 (L_1 - M) L_1 L_1$$

$$L_1 L_1 (L_2 - M) L_2$$

$$= (L_1 + L_2) - 2M - (2)$$

$$L_1 L_1 (L_1 - M) L_1$$

equating (1) and (2)

$$(L_2 - M) L_2 L_1 (L_2 - M) L_2$$

$$(L_1 + L_2) = (L_1 + L_2)$$

$$(L_1 - M) L_1 L_1 (L_1 - M) L_1$$

$$L_1 L_2 - L_2$$

$$L = L_1 + L_2 - 2M - (3)$$

$$L_1 + L_2 - 2M$$

Equation 3 is when mutual field assists the separate fields.

For when the two fields oppose each other

$$L_1 L_2 - L_2$$

$$L =$$

$$L_1 + L_2 + 2M$$

Example: Two coils of inductances 4 and 6 henry are connected in parallel. If their mutual inductance is 3 henry, calculate the equivalent inductance of the combination if (i) mutual

inductance assists the self-inductance (ii) mutual inductance opposes the self-inductance

2.15 ENERGY STORED

As with an electric field, a magnetic field also stores energy. When the current through an inductive circuit is interrupted, by opening a switch, this energy is released. This is the reason why a spark or arc occurs between the contacts of the switch, when it is opened.

Consider an inductor connected in a circuit, in which the current increases uniformly, to some steady value I amp. This current change is illustrated in Fig. 2. The magnitude of the emf induced by this change of current is given by

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□□

□ = □□□□

□

Figure 5

The average power input to the coil during this time is:

□□□□□□ □□□□□ = □ × □□□□□□ □□□□□□

From the graph, it may be seen that the average current over this time is $I/2$ amp.

Therefore,

1 □□□ □□ 2

□□□□□□ □□□□□ = □□ = =

2 2□ 2□

But, energy stored = average power × time

□□ 2

□□□□□□ □□□□□□ □ = ×□

2□

1

□ = □□ 2 □□□□□□ - - - - - (1)

2

Equation 1 applies to a single inductor. When two coils possess mutual inductance, and are connected in series, both will store energy. In this situation, the total energy stored is given by the equation

1 1

□ = □1 □12 + □2 □22 ± □□1 □2 □□□□□□

2 2

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When the coils are connected in series such that the two fluxes produced act in the same direction, the total flux is increased and the coils are said to be connected in series aiding. In

this case the total energy stored in the system will be increased, so the last term in the above equation is added, i.e. the positive sign applies. If, however, the connections to one of the coils are reversed, then the two fluxes will oppose each other, the total flux will be reduced, and the coils are said to be in series opposition. In this case the negative sign is used. These two connections are shown in Figure .

Example: Two inductors, of inductance 25 mH and 40mH respectively, are wound on a common ferromagnetic core, and are connected in series with each other. The coupling coefficient, k , between them is 0.8. When the current flowing through the two coils is 0.25 A, calculate (a) the energy stored in each, (b) the total energy stored when the coils are connected (i) in series aiding, and (ii) in series opposition.

2.16 AUTOMOBILE IGNITION SYSTEM

Practical application of mutual induction is found in the single-spark petrol-engine ignition system extensively employed in automobiles and air engines. Fig.7 shows the circuit diagram of such a system as applied to a 4-cylinder automobile engine. It has a spark coil (or induction coil) which consists of a primary winding (of a few turns) and a secondary winding (of a large number of turns) wound on a common iron core (for increasing mutual induction). The primary circuit (containing battery B) includes a 'make and break contact' actuated by a timer cam. The secondary circuit includes the rotating blade of the distributor and the spark gap. The timer cam and the distributor are mounted on the same shaft and are geared to rotate at exactly half the speed of the engine shaft. It means that in the case of automobile engines (which are fourcycle engines) each cylinder is fired only once for every two revolutions of the engine shaft

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Figure 7

When timer cam rotates, it alternately closes and opens the primary circuit. During the time primary circuit is closed, current through it rises exponentially, and so does the magnetic field of the primary winding. When the cam suddenly opens the primary circuit, the magnetic field collapses rapidly thereby producing a very large e.m.f. in secondary by mutual induction. During the time this large e.m.f. exists, the distributor blade rotates and connects the secondary winding across the proper plug and so the secondary circuit is completed except for the spark gap in the spark plug. However, the induced e.m.f. is large enough to make the current jump across the gap thus producing a spark which ignites the explosive mixture in the engine cylinder. The function of capacitor C connected across the 'make and break' contact is two-fold :

- (i) to make the break rapid so that large e.m.f. is induced in secondary and
- (ii) to reduce sparking and burning at the 'make-and-break' contact thereby prolonging their life.

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Chapter 3

A.C. FUNDAMENTALS

This chapter deals with the concepts, terms and definitions associated with alternating quantities. The term alternating quantities refers to any quantity (current, voltage, flux, etc.), whose polarity is reversed alternately with time. For convenience, they are commonly referred to as a.c. quantities.

On completion of this chapter you should be able to:

- i. Explain the method of producing an a.c. waveform.
- ii. Define all of the terms relevant to a.c. waveforms.
- iii. Obtain values for an a.c., both from graphical information and when expressed in mathematical form.
- iv. Understand and use the concept of phase angle.
- v. Use both graphical and phasor techniques to determine the sum of alternating quantities.

3.1 INTRODUCTION

Alternating voltage may be generated by rotating a coil in a magnetic field, as shown in Fig. 1 (a) or by rotating a magnetic field within a stationary coil, as shown in Fig. 1 (b).

Figure 1

The value of the voltage generated depends, in each case, upon the number of turns in the coil, strength of the field and the speed at which the coil or magnetic field rotates. Alternating voltage may be generated in either of the two ways shown above, but rotating-field method is the one which is mostly used in practice.

3.2 EQUATIONS OF THE ALTERNATING VOLTAGES AND CURRENTS

Consider a rectangular coil, having N turns and rotating in a uniform magnetic field, with an angular velocity of ω radian/second, as shown in Fig. 2. Let time be measured from the X-

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axis. Maximum flux Φ_m is linked with the coil, when its plane coincides with the X-axis. In time t seconds, this coil rotates through an angle $\theta = \omega t$. In this deflected position, the component of the flux which is perpendicular to the plane of the coil, is $\Phi = \Phi_m \cos \omega t$. Hence, flux linkages of the coil at any time are $N \Phi = N \Phi_m \cos \omega t$. According to Faraday's Laws of Electromagnetic Induction, the e.m.f. induced in the coil is given by the rate of change of flux-linkages of the coil. Hence, the value of the induced e.m.f. at this instant (i.e. when $\theta = \omega t$) or the instantaneous value of the induced e.m.f. is

Figure 2

Similarly, the equation of induced alternating current is $i = I_m \sin \omega t$... (iv) provided the coil circuit has been closed through a resistive load. Since $\omega = 2\pi f$, where f is the frequency of rotation of the coil, the above equations of the voltage and current can be written as

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Figure 3

3.3 SIMPLE WAVEFORMS

The shape of the curve obtained by plotting the instantaneous values of voltage or current as the ordinate against time as abscissa is called its waveform or wave-shape.

Figure 4

An alternating voltage or current may not always take the form of a systematical or smooth wave such as that shown in Fig. 3. Thus, Fig. 4 also represents alternating waves. But while it is scarcely possible for the manufacturers to produce sine-wave generators or alternators, yet sine wave is the ideal form sought by the designers and is the accepted standard. The waves deviating from the standard sine wave are termed as distorted waves. In general, however, an alternating current or voltage is one the circuit direction of which reverses at regularly recurring intervals.

Time Period: The time taken by an alternating quantity to complete one cycle is called its time period T . For example, a 50-Hz alternating current has a time period of $1/50$ second.

Frequency: The number of cycles/second is called the frequency of the alternating quantity. Its unit is hertz (Hz).

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$f = PN/120$ where N = revolutions in r.p.m. and P = number of poles

Amplitude: The maximum value, positive or negative, of an alternating quantity is known as its amplitude.

Phase: By phase of an alternating current is meant the fraction of the time period of that alternating current which has elapsed since the current last passed through the zero position of reference.

Root-Mean-Square (R.M.S.) Value

The r.m.s. value of an alternating current is given by that steady (d.c.) current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time. It is also known as the effective or virtual value of the alternating current, the former term being used more extensively.

The standard form of a sinusoidal alternating current is $i = I_m \sin \omega t = I_m \sin \theta$. The mean of the squares of the instantaneous values of current over one complete cycle is (even the value over half a cycle will do).

Hence, we find that for a symmetrical sinusoidal current

$$I_{\text{R.M.S.}} = \frac{I_m}{\sqrt{2}} = 0.707 \times I_m$$

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The r.m.s. value of an alternating current is of considerable importance in practice, because the ammeters and voltmeters record the r.m.s. value of alternating current and voltage respectively. In electrical engineering work, unless indicated otherwise, the values of the given current and voltage are always the r.m.s. values

Average Value

The average value of an alternating current is expressed by that steady current which transfers across any circuit the same charge as is transferred by that alternating current during the same time. In the case of a symmetrical alternating current (i.e. one whose two half-cycles are exactly similar, whether sinusoidal or non-sinusoidal), the average value over a complete cycle is zero. Hence, in their case, the average value is obtained by adding or integrating the instantaneous values of current over one half-cycle only. But in the case of an unsymmetrical alternating current (like half-wave rectified current) the average value must always be taken over the whole cycle.

[illegible]

Note. R.M.S. value is always greater than average value except in the case of a rectangular wave when both are equal.

Form Factor

It is defined as the ratio, $K_f = \text{r.m.s. value} / \text{average value}$

0.701□□
□□ = = 1.11
0.637□□

Crest or Peak or Amplitude Factor

It is defined as the ratio $K_a = \text{maximum value} / \text{rms value}$

$$\frac{1}{\sqrt{2}} = \frac{1}{1.414} = 0.707$$

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Example 1: An alternating current of frequency 60 Hz has a maximum value of 120 A. Write down the

equation for its instantaneous value. Reckoning time from the instant the current is zero and is becoming

positive, find (a) the instantaneous value after $1/360$ second and (b) the time taken to reach 96 A for the first time.

Solution.

The instantaneous current equation is $i = 120 \sin 2\pi ft = 120 \sin 120\pi t$

Now when $t = 1/360$ second, then

(a)

1

$$i = 120 \sin (120 \times \frac{1}{360} \times \pi) \dots \text{instantaneous value}$$

360

1

$$= 120 \sin (120 \times 180 \times \frac{1}{360} \times \pi) \dots \text{instantaneous value}$$

360

$$= 120 \sin 60^\circ = 103.9 \text{ A}$$

(b)

$$96 = 120 \times \sin 2\pi \times 180 \times 60 \times t \dots \text{instantaneous value}$$

96

$$\sin(360 \times 60 \times t) = 0.8$$

120

$$360 \times 60 \times t = \sin^{-1} 0.8 = 53^\circ \text{ (angle)}$$

$$\therefore t = \frac{53}{360 \times 60} = 53/360 \times 60 = 0.00245 \text{ seconds.}$$

Example 2: An alternating current varying sinusoidally with a frequency of 50 Hz has an RMS value of 20 A. Write down the equation for the instantaneous value and find this value (a) 0.0025 second (b) 0.0125 second after passing through a positive maximum value. At what time, measured from a positive maximum value, will the instantaneous current be 14.14 A?

Example 3: A rectangular coil, measuring 25 cm by 20 cm, has 80 turns. The coil is rotated, about an axis parallel with its longer sides, in a magnetic field of density 75 mT. If the speed of rotation is 3000 rev/min, calculate, from first principles, (a) the amplitude, r.m.s. and average values of the emf, (b) the frequency and period of the generated waveform, (c) the instantaneous value, 2ms after it is zero.

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3.4 PHASE DIFFERENCE

When two alternating quantities of the same frequency have different zero points, they are said to have a phase difference. The angle between the zero points is the angle of phase difference.

In Phase

Two waveforms are said to be in phase, when the phase difference between them is zero. That is the zero points of both the waveforms are same. The waveform, phasor and equation representation of two sinusoidal quantities which are in phase is as shown. The figure shows that the voltage and current are in phase.

$$i = I_m \sin \omega t$$

$$i = I_m \sin \omega t$$

Lagging

In the figure shown, the zero point of the current waveform is after the zero point of the voltage waveform. Hence the current is lagging behind the voltage. The waveform, phasor and equation representation is as shown.

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$$i = I_m \sin \omega t$$

$$i = I_m \sin \omega t - \phi$$

Lagging

In the figure shown, the zero point of the current waveform is after the zero point of the voltage waveform. Hence the current is lagging behind the voltage. The waveform, phasor and equation representation is as shown.

$$i = I_m \sin \omega t$$

$$i = I_m \sin \omega t + \phi$$

3.5 PHASOR REPRESENTATION OF ALTERNATING QUANTITIES

A phasor is a rotating vector. Apart from the fact that a phasor rotates at a constant velocity, it has exactly the same properties as any other vector. Thus its length corresponds to the magnitude of a quantity. It has one end arrowed, to show the direction of action of the quantity.

Consider two such rotating vectors, v_1 and v_2 , rotating at the same angular velocity, rad/s. Let them rotate in a counter clockwise direction, with v_2 lagging behind v_1 by $\pi/6$ radian (30°).

This situation is illustrated in Fig.7 .

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Figure 7

Notes

- Any a.c. quantity can be represented by a phasor, provided that it is a sine wave.
- Any number of a.c. voltages and/or currents may be shown on the same phasor diagram, provided that they are all of the same frequency.
- Figure 7 shows a counter-clockwise arrow, with ω rad/s. This has been shown here to emphasise the point that phasors must rotate in this direction only. It is normal practice to omit this from the diagram.
- When dealing with a.c. circuits, r.m.s. values are used almost exclusively. In this case, it is normal to draw the phasors to lengths that correspond to r.m.s. values.

Example 4: Four currents are as shown below. Draw to scale the corresponding phasor diagram.

$$i_1 = 2.5 \sin(\omega t + \pi/4) \text{ A}, i_2 = 4 \sin(\omega t - \pi/3) \text{ A}$$

$$i_3 = 6 \sin \omega t \text{ amp}, i_4 = 3 \cos \omega t \text{ amp}$$

Addition of Alternating Quantities

Consider two alternating currents, $i_1 = I_{m1} \sin \omega t$ amp and $i_2 = I_{m2} \sin (\omega t + \pi/4)$ amp, that are to be added together. There are three methods of doing this, as listed below.

> Plotting them on graph paper. Their ordinates are then added together, and the resultant waveform plotted. This is illustrated in Fig. 8. The amplitude, I_m , and the phase angle, θ , of the resultant current are then measured from the two axes.

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> Drawing a scaled phasor diagram, as illustrated in Fig.9. The resultant is found by completing the parallelogram of vectors. The amplitude and phase angle are then measured on the diagram.

> Resolving the two currents, into horizontal and vertical components, and applying Pythagoras theorem. This method involves using a sketch of the phasor diagram, followed by a purely mathematical process. This phasor diagram, including the identification of the horizontal and vertical components, is shown in Fig. 9.

The triangle of H.C., V.C., and the resultant current, is shown in Fig.10 . From this, we can apply Pythagoras theorem to determine the amplitude and phase angle, thus

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Figure 10

The graphical technique is very time-consuming (even for the addition of only two quantities). The accuracy also leaves much to be desired, a phasor diagram, drawn to scale, can be the quickest method of solution. However, it does require considerable care, in order to ensure a reasonable degree of accuracy, the use of the resolution of phasors is, with practice, a rapid technique, and yields a high degree of accuracy. Unless specified otherwise it is the technique you should use.

Example 5: Determine the phasor sum of the two voltages specified below.

$$V_1 = 25 \sin(\omega t + \pi/2)$$

$$V_2 = 15 \sin(\omega t - \pi/6)$$

Example 6: Calculate the phasor sum of the three currents listed below.

$$i_1 = 6 \sin \omega t$$

$$i_2 = 8 \sin(\omega t - \pi/2)$$

$$i_3 = 4 \sin(\omega t + \pi/6)$$

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3.6 A.C. THROUGH RESISTANCE, INDUCTANCE AND CAPACITANCE

A.C. Through Pure Ohmic Resistance Alone

The circuit is shown in Fig. 13. Let the applied voltage be given by the equation.

$$v = V_m \sin \omega t$$

Let R = ohmic resistance; i = instantaneous current Obviously, the applied voltage has to supply ohmic voltage drop only. Hence for equilibrium $v = iR$;

$$v = V_m \sin \omega t = iR$$

$$i = \frac{v}{R}$$

$$i = \frac{V_m}{R} \sin \omega t$$

$$I_m = \frac{V_m}{R}$$

$$I = \frac{V_m}{R} \sin \omega t$$

$$I = \frac{V}{R} \sin \omega t$$

$$I = \frac{V}{R} \sin \omega t$$

Hence,

$$i = \frac{v}{R} \sin \omega t$$

Instantaneous power, $p = vi = \frac{v^2}{R} = \frac{V_m^2}{R} \sin^2 \omega t$

$$p = \frac{V_m^2}{R} \sin^2 \omega t$$

$$\frac{V_m^2}{R} \sin^2 \omega t = \frac{V_m^2}{R} (1 - \cos 2\omega t)$$

$$2$$

Average value of $\cos 2\omega t$ is zero

$$\frac{V_m^2}{R} \sin^2 \omega t = \frac{V_m^2}{R} (1 - \cos 2\omega t)$$

$$P = \frac{V_m^2}{R} \times \frac{1}{2} = \frac{V_m^2}{2R}$$

$$2 \sqrt{2} \sqrt{2}$$

Where V = r.m.s. value of applied voltage.

I = r.m.s. value of the current.

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Example 6

An ac circuit consists of a pure resistance of 10Ω and is connected to an ac supply of 230 V, 50 Hz. Calculate the (i) current (ii) power consumed and (iii) equations for voltage and current

Solution

i.

$$V = 230$$

$$R = 10$$

$$I = \frac{V}{R} = \frac{230}{10} = 23 \text{ A}$$

ii.

$$P = I^2 R = 23^2 \times 10 = 5260 \text{ W}$$

iii.

$$V_m = V\sqrt{2} = 230\sqrt{2} = 325.27$$

$$I_m = I\sqrt{2} = 23\sqrt{2} = 32.52$$

$$\omega = 2\pi f = 2 \times 50 \times 3.14 = 314 \text{ rad/s}$$

$$v = 325.27 \sin 314t \text{ V}$$

$$i = 32.52 \sin 314t$$

Example 7: A 60-Hz voltage of 115 V (r.m.s.) is impressed on a 100 ohm resistance : (i) Write the time equations for the voltage and the resulting current. Let the zero point of the voltage wave be at $t = 0$ (ii) Show the voltage and current on a time diagram. (iii) Show the voltage and current on a phasor diagram. Answer $v(t) = 1.63 \sin 377 t$ and $i(t) = 1.63 \sin 377 t$

A.C. Through Pure Inductance Alone

Whenever an alternating voltage is applied to a purely inductive coil, a back e.m.f. is produced due to the self-inductance of the coil. The back e.m.f., at every step, opposes the rise or fall of current through the coil. As there is no ohmic voltage drop, the applied voltage has to overcome this self-induced e.m.f. only. So at every step

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Let

$$v = V_m \sin \omega t$$

and

$$i = I_m \sin(\omega t - \phi)$$

Then

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

or

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

$$= I_m \sin(\omega t - \phi)$$

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

or

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

or

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

or

$$V_m \sin \omega t = I_m \sin(\omega t - \phi)$$

Where

V_m

$$V_m =$$

I_m

Instantaneous power

The instantaneous power in the above circuit can be derived as follows

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As seen from the above equation, the instantaneous power is fluctuating in nature.

Average power

From the instantaneous power we can find the average power over one cycle as follows

The average power in a pure inductive circuit is zero. Or in other words, the power consumed by a pure inductance is zero.

The voltage, current and power waveforms of a purely inductive circuit is as shown in the figure.

Example: A pure inductive coil allows a current of 10A to flow from a 230V, 50 Hz supply. Find (i) inductance of the coil (ii) power absorbed and (iii) equations for voltage and current.

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Solution

i.

$$V = 230$$

$$I = 10 = 23\Omega$$

$$I = 10$$

$$I = 10$$

$$I = 0.073$$

$$2I = 20$$

ii. $P=0$

iii.

$$V_m = V/\sqrt{2} = 325.27$$

$$I_m = I/\sqrt{2} = 14.14$$

$$\omega = 2\pi f = 314 \text{ rad/s}$$

$$V = 325.27 \sin 314t$$

$$I = 14.14 \sin(314t - \pi/2)$$

A.C. Through Pure Capacitance Alone

When an alternating voltage is applied to the plates of a capacitor, the capacitor is charged first in one direction and then in the opposite direction.

$$V = V_m \sin \omega t$$

$$I = \frac{V}{X_c} = \frac{V_m \sin \omega t}{X_c}$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$$I = \frac{V_m}{X_c} \sin \omega t$$

$i = I_m \sin(\omega t + \phi)$

$i = I_m \sin(\omega t + 2)$

$i = I_m \sin(\omega t + \phi)$

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The denominator $X_C = 1/\omega C$ is known as capacitive reactance and is in ohms.

The maximum value of the instantaneous power is $V_m I_m / 2$.

A $318 \mu\text{F}$ capacitor is connected across a 230V, 50 Hz system. Find (i) the capacitive reactance (ii) rms value of current and (iii) equations for voltage and current.

Solution

i.

$X_C = 1 / \omega C$

$X_C = 1 / (2 \pi \times 50 \times 318 \times 10^{-6})$

$X_C = 10 \Omega$

ii.

$V = 230$

$I = V / X_C = 23 / 10$

$I = 2.3$

iii.

$V = V_m \sin \omega t = 325.27 \sin 314 t$

$I = I_m \sin \omega t = 32.53 \sin 314 t$

$V = 2 I = 314 \times 2.3 / 314$

$V = 325.27 \sin 314 t$

$I = 32.53 \sin(314 t + \pi/2)$

Example: A 50-Hz voltage of 230 volts effective value is impressed on a capacitance of $26.5 \mu\text{F}$. (a) Write the time equations for the voltage and the resulting current. Let the zero axis of the voltage wave be at $t = 0$. (b) Show the voltage and current on a time diagram. (c) Show the voltage and current on a phasor diagram.

Answer a) $v(t) = 325 \sin 314 t$; $i(t) = 2.71 \sin(314 t + \pi/2) = 2.71 \cos 314 t$.

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A.C. Through Resistance and Inductance

A pure resistance R and a pure inductive coil of inductance L are shown connected in series in Fig. 14

Figure 14

$Z = \sqrt{R^2 + X_L^2} = \sqrt{(R)^2 + (\omega L)^2} = \sqrt{R^2 + \omega^2 L^2}$

$Z = \sqrt{R^2 + \omega^2 L^2}$

$Z = \sqrt{R^2 + \omega^2 L^2}$

$$\sqrt{R^2 + X_L^2}$$

The quantity $\sqrt{R^2 + X_L^2}$ is known as the impedance (Z) of the circuit.

$$\tan \phi = \frac{X_L}{R}$$

$$\cos \phi = \frac{R}{Z}$$

$$\sin \phi = \frac{X_L}{Z}$$

The mean power consumed by the circuit is given by the product of V and that component of the current I which is in phase with V.

So

$$P = V \times I \cos \phi$$

The term 'cos ϕ ' is called the power factor of the circuit.

Active, Reactive and Apparent Power

Let a series R-L circuit draw a current of I when an alternating voltage of r.m.s. value V is applied to it. Suppose that current lags behind the applied voltage by ϕ . The three powers drawn by the circuit are as under:

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i. Apparent power (S) It is given by the product of r.m.s. values of applied voltage and circuit current. $\therefore S = VI = (IZ)I = I^2Z$ volt-amperes (VA)

ii. Active power (P or W) It is the power which is actually dissipated in the circuit resistance. $P = I^2R = VI \cos \phi$ watts

iii. Reactive power (Q) It is the power developed in the inductive reactance of the circuit. $Q = I^2X_L = I^2Z \sin \phi = I.(IZ) \sin \phi = VI \sin \phi$ volt-amperes-reactive (VAR)

These three powers are shown in the power triangle of Fig 1 from where it can be seen that $S^2 = P^2 + Q^2$.

Q-factor of a Coil

Reciprocal of power factor is called the Q-factor of a coil or its figure of merit. It is also known as quality factor of the coil.

$$Q \text{ factor} = 1/\text{power factor OR } 1/\cos \phi$$

$$\text{Also } Q = 2\pi (\text{maximum energy stored})/(\text{energy dissipated per cycle})$$

Example : In a series circuit containing pure resistance and a pure inductance, the current and the voltage are expressed as : $i(t) = 5 \sin(314t + 2\pi/3)$ and $v(t) = 15 \sin(314t + 5\pi/6)$ (a) What is the impedance of the circuit ? (b) What is the value of the resistance? (c) What is the inductance in henrys? (d) What is the average power drawn by the circuit? (e) What is the power factor?

Solution.

$$\phi_{\text{current}} - \phi_{\text{voltage}} = 2\pi/3 - 5\pi/6 = 2 \times 180^\circ/3 - 5 \times 180^\circ/6 = 120^\circ - 150^\circ = -30^\circ$$

$$\phi_{\text{voltage}} - \phi_{\text{current}} = 5\pi/6 - 2\pi/3 = 5 \times 180^\circ/6 - 2 \times 180^\circ/3 = 150^\circ - 120^\circ = 30^\circ$$

Hence, current lags behind voltage by 30° . It means that it is an R-L circuit.

$$I = \frac{V}{Z} = 3 \text{ A}$$

$$\cos 314 = 2 \times 10^3 \text{ V} = 50 \text{ V}$$

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$$I = \frac{V}{Z} \cos 30^\circ = 2.6 \text{ A} ; I = \frac{V}{Z} \cos 30^\circ = 1.5$$

$$I = 2 \times 10^3 = 314 \text{ V} = 1.5, I = 4.78 \text{ V}$$

2

2

5

$$I = \frac{V}{Z} = () \times 2.6 = 32.5 \text{ A}$$

$$\sqrt{2}$$

$$\cos 30^\circ = 0.866 \text{ (approx.)}$$

R-C Series circuit

Consider an AC circuit with a resistance R and a capacitance C connected in series as shown in the figure (a).

The alternating voltage v is given by

$$v = V_m \sin \omega t$$

The current flowing in the circuit is i. The voltage across the resistor is VR and that across the capacitor is VC.

VR = IR is in phase with I

VC = IXC lags behind the current by 90 degrees

The current I is taken as the reference phasor. The voltage VR is in phase with I and the voltage VC lags behind the current by 90°. The resultant voltage V can be drawn as shown in the figure. From the phasor diagram we observe that the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ.

The waveform and equations for an RC series circuit can be drawn as below.

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$$v = V_m \sin \omega t$$

$$i = I_m \sin \omega t + \phi$$

From the phasor diagram, the expressions for the resultant voltage V and the angle can be derived as follows.

$$V = \sqrt{V_R^2 + V_C^2}$$

$$V_R = I R, V_C = I X_C$$

$$V = \sqrt{I^2 R^2 + I^2 X_C^2}$$

$$V = I \sqrt{R^2 + X_C^2}$$

$$I = \frac{V}{Z}$$

Where the impedance Z is

$$Z = R^2 + X_C^2$$

The phase angle

$$\Phi = \tan^{-1} \left(\frac{X_C}{R} \right)$$

$$\Phi = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

$$\Phi = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

$$\Phi = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

$$\Phi = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

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$$\Phi = \tan^{-1} \left(\frac{1}{\omega RC} \right)$$

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Average power

Hence the power in an RC series circuit is consumed only in the resistance. The capacitance does not consume any power.

Impedance Triangle

We can derive a triangle called the impedance triangle from the phasor diagram of an RC series circuit as shown

Phasor algebra for RC series circuit

Example: A Capacitor of capacitance $79.5\mu\text{F}$ is connected in series with a non-inductive resistance of 30Ω across a 100V , 50Hz supply. Find (i) impedance (ii) current (iii) phase angle (iv) Equation for the instantaneous value of current.

Solution

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R-L-C Series circuit

Consider an AC circuit with a resistance R , an inductance L and a capacitance C connected in series as shown in the figure. The alternating voltage v is given by

The current flowing in the circuit is i . The voltage across the resistor is V_R , the voltage across the inductor is V_L and that across the capacitor is V_C .

$V_R = IR$ is in phase with I

$V_L = IXL$ leads the current by 90 degrees

$V_C = IXC$ lags behind the current by 90 degrees

With the above information, the phasor diagram can be drawn as shown. The current I is taken as the reference phasor. The voltage V_R is in phase with I , the voltage V_L leads the current by

90° and the voltage V_C lags behind the current by 90° . There are two cases that can occur

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$V_L > V_C$ and $V_L < V_C$ depending on the values of X_L and X_C . And hence there are two possible phasor diagrams. The phasor $V_L - V_C$ or $V_C - V_L$ is drawn and then the resultant voltage V is drawn

From the phasor diagram we observe that when $V_L > V_C$, the voltage leads the current by an angle Φ or in other words the current lags behind the voltage by an angle Φ . When $V_L < V_C$, the voltage lags behind the current by an angle Φ or in other words the current leads the voltage by an angle Φ .

From the phasor diagram, the expressions for the resultant voltage V and the angle Φ can be derived as follows.

Where impedance

Phase angle

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From the expression for phase angle, we can derive the following three cases

Case (i): When $X_L > X_C$

The phase angle Φ is positive and the circuit is inductive. The circuit behaves like a series RL circuit.

Case (ii): When $X_L < X_C$

The phase angle Φ is negative and the circuit is capacitive. The circuit behaves like a series RC circuit.

Case (iii): When $X_L = X_C$

The phase angle $\Phi = 0$ and the circuit is purely resistive. The circuit behaves like a pure resistive circuit.

The voltage and the current can be represented by the following equations. The angle Φ is positive or negative depending on the circuit elements.

Average power

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Hence the power in an RLC series circuit is consumed only in the resistance. The inductance and the capacitance do not consume any power.

Phasor algebra for RLC series circuit

Example 10: A 230 V, 50 Hz ac supply is applied to a coil of 0.06 H inductance and 2.5 resistance connected in series with a 6.8 μF capacitor. Calculate (i) Impedance (ii) Current (iii) Phase angle between current and voltage (iv) power factor (v) power consumed

Example 11: A resistance R , an inductance $L = 0.01 \text{ H}$ and a capacitance C are connected in series. When an alternating voltage $v = 400\sin(3000t - 20^\circ)$ is applied to the series combination, the current flowing is $10\sqrt{2}\sin(3000t - 65^\circ)$. Find the values of R and C .

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Example 12: A coil of pf 0.6 is in series with a $100\mu\text{F}$ capacitor. When connected to a 50Hz supply, the potential difference across the coil is equal to the potential difference across the capacitor. Find the resistance and inductance of the coil.

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Resonance in R-L-C Circuits

The frequency at which the net reactance of the series circuit is zero is called the resonant frequency f_0 . Its value can be found as under :

$$X_L - X_C = 0 \text{ or } X_L = X_C \text{ or } \omega_0 L = 1/\omega_0 C$$

1

$$\omega_0 =$$

$$2\sqrt{\frac{L}{C}}$$

When a series R-L-C circuit is in resonance, it possesses minimum impedance $Z = R$. Hence, circuit current is maximum, it being limited by value of R alone. The current $I_0 = V/R$ and is in phase with V .

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